A gear train is arranged between the main spindle and the driven shaft when indexing is done by differential indexing method. The work is rotated as usual when the handle is rotated. At the same time, the index plate is also made to rotate a small amount through the gear train when indexing is by this differential indexing method, the index plate is released from the lock pin.

5.19 Indexing methods

There are several methods of indexing and they are

1. Direct or rapid indexing
2. Plain or simple indexing
3. Compound indexing
4. Differential indexing
5. Angular indexing

5.20 Safety precautions

Before operating the milling machine, the operator should know how to operate various controls of the machine. It should be ensured that the workpiece is held rigidly on the milling machine table. The cutter should not be in contact with the work even before the machining is commenced.

The spindle speed of machine should not be altered when the machine is in operation. When the power of the machine table is on the arbor should not be removed or tightened. The operator should keep his body away from the rotating cutter. No steps should be taken to measure the workpiece while the cutter is cutting or revolving near the workpiece when the machine is in operation, safety guards should be placed in their position to prevent coolant and metal chips from spilling out.

The metal chips should be removed with suitable brushes and with bare hands. The operation should seek assistance from others while handing special attachments and heavy workpieces. The operator should always be present in person at the machine tool when the machine is in operation.

The machine tool should always be started and stopped by the operator himself. Danger's can be averted by handling the cutters with sharp cutting edges with great care. The machine tool should be kept clean. Milling cutters and measuring instruments should not be placed on the machine. The attention of the operation should always be focused on the task only. When troubles happen in the machine, they should be corrected with assistance of proper technicians in general safety should be ensured to the operator, the workpiece and the cutting tool.

ACTIVITIES

1. To arrange the students visit nearby industry and demonstrate the milling machine construction details and various kinds of operations.
2. To explain the different types of indexing methods with practical examples.

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